```
#include <stdio.h>
#include <stdlib.h>

int mutex = 1;
int full = 0;
int empty = 5;
int x = 0;
int buffer[5];
int in = 0;
int out = 0;
int wait(int s) {
   return (--s);
}
int signal(int s) {
   return (++s);
}
void producer() {
   if (mutex == 1 && empty != 0) {
      mutex = wait(mutex);
      full = signal(full);
      empty = wait(empty);
      x++;
      buffer[in] = x;
      printf("\nProducer has produced %d : item %d", in, x);
      in = (in + 1) % 5;

      mutex = signal(mutex);
   } else {
      printf("\nBuffer is full!");
   }
}
```

```
void consumer() {
    if (mutex == 1 && full != 0) {
        mutex = wait(mutex);
        full = wait(full);
        empty = signal(empty);
        int item = buffer[out];
        printf("\nConsumer has consumed %d : item %d", out, item);
        out = (out + 1) % 5;
        mutex = signal(mutex);
    } else {
        printf("\nBuffer is empty!");
    }
}

int main() {
    int n;
    printf("\n1. Producer\n2. Consumer\n3. Exit");
    while (1) {
        printf("\n\nEnter your choice: ");
        scanf("%d", &n);
        switch (n) {
            case 1:
                producer();
                break;
            case 2:
                consumer();
                break;
            case 3:
                printf("Exiting program...\n");
                exit(0);
```

```
            default:
                printf("Invalid choice!");
        }
    }
    return 0;
}
```

OUTPUT :

```
Enter your choice: 1
Producer has produced 0 : item 1
Enter your choice: 1
Producer has produced 1 : item 2
Enter your choice: 1
Producer has produced 2 : item 3
Enter your choice: 2
Consumer has consumed 0 : item 1
Enter your choice: 2
Consumer has consumed 1 : item 2
Enter your choice: 2
Consumer has consumed 2 : item 3
Enter your choice: 2
Buffer is empty!
```